Muhammad Awais

Nationality: Pakistani | **Place of birth:** Dera Ismail Khan | **Gender:** Male

Email: mawais2500@alumno.ipn.mx | **Affiliation:** ESFM – IPN, Mexico

Motivation

- Practical expertise in **Bayesian inference and MCMC techniques**, with extensive use of **MontePython** for cosmological parameter estimation.
- Experience building and modifying Boltzmann solvers, including **CLASS** and **CAMB**, for precision cosmology and beyond-ΛCDM models.
- I am highly motivated to deepen my expertise in **cosmological data analysis**, **statistical methods**, and **theoretical modeling** through this workshop unique structure spanning training, observations, and theory.

Education

PhD in Physics *2025–Present*
ESFM–IPN, Mexico City, Mexico

- Currently working on dark matter and dark energy models using observational data from **DESI**.
- Focus on cosmological parameter estimation and large-scale structure analysis.
- Website: https://www.esfm.ipn.mx/

MS / MPhil in Physics *Sep 2021–Jun 2023*
COMSATS University Islamabad (CUI), Pakistan

- Thesis: Calculation of Dark Matter Relic Density in the Non-Holomorphic Supersymmetric Standard Model.
- CGPA: 3.70 / 4.00
- Website: https://www.comsats.edu.pk/

Professional Experience

University Research Assistant *2025–Present*
ESFM–IPN, Mexico City, Mexico

- Development of reproducible **Bayesian inference workflows** for cosmological parameter estimation.
- Analysis of large-scale structure data using Python-based scientific computing tools.

Practical Exposure

Teaching Assistant – Advanced Cosmology *Spring Semester 2026 (current)*
ESFM–IPN, Mexico City, Mexico
Main Responsibilities:

- Assist in teaching advanced cosmology topics, including the accelerated expansion of the Universe, Boltzmann equations in an expanding background, initial conditions, and cosmological perturbations.
- Support computational sessions using **CLASS** and **MontePython** for cosmological parameter estimation.

- Guide students in practical Python applications in cosmology, including **Astropy**-based data analysis.
- Contribute to discussions on inhomogeneities, anisotropies, the matter power spectrum, CMB physics, and Baryon Acoustic Oscillations (BAO).

Ongoing Projects

Constraining Scalar Field Dark Matter Using Baryon Acoustic Oscillations from DESI *2025–Current*

Main Contributions:

- Explore **scalar field dark matter (SFDM)** as an alternative to **ΛCDM** using precision cosmological probes.
- Use high-precision **DESI** large-scale structure measurements to constrain SFDM parameter space through **BAO** and related clustering statistics.
- Perform end-to-end **computational analysis** workflows: catalog preparation, quality checks, and handling calibration/selection effects for robust inference.
- Run **Bayesian inference with MCMC** (e.g., MontePython + CLASS/CAMB) to obtain posterior constraints and assess model viability.

Galaxy Evolution in Different Cosmic Web Environments: A T-Web Analysis of LRG, ELG, and BGS Samples *2025–Current*

Main Contributions:

- Perform **Tidal Tensor (T-Web)** cosmic web classification using **DESI DR1** galaxy samples (LRG, ELG, BGS) to identify voids, sheets, filaments, and knots across multiple smoothing scales.
- Reconstruct the large-scale matter density field on a 3D grid, compute tidal tensor eigenvalues, and analyze the **scale dependence** of environment fractions across tracers.
- Apply mass-dependent red/blue galaxy separation (Otsu method) and quantify environmental quenching using red/blue fraction statistics across stellar-mass and redshift bins.
- Build graph-based representations of the cosmic web and apply **Graph Neural Networks (GNNs)** to learn environmental features and compare observational trends with simulation predictions.

Technical Skills

Programming & Computing:	Python (scientific computing), C++ (intermediate), Linux, LaTeX
Scientific / HEP tools:	MontePython, CLASS, ROOT, SARAH, SPheno, MicroOmega, Mathematica, Pythia, FormCalc/LoopTools

Conferences

- Mexican Astro-Cosmo Statistics School, University of Guanajuato, León, Mexico (2025).
- XIII Majorana Lecture Series — Edward Witten, University of Naples Federico II, Italy (2024).
- School on LHC Physics, National Centre for Physics (NCP), Pakistan (2023).
- Modern Trends in High Energy Physics, The Islamia University of Bahawalpur, Pakistan (2023).
- Workshop on Astronomy, Telescopes, and Astrophysics, NUST Islamabad, Pakistan (2022).
- Joint ICTP–IAEA International School on Radionuclide Migration, ICTP Trieste (Remote) (2022).
- 2nd Dr. Riaz-ud-Din Memorial School on Physics, UAF, Pakistan (2022).

International Courses

- Explore Einstein's Theories of Relativity Using Wolfram.
  coursera.org/verify/A86E5LDDNBC9
- Understanding Modern Physics I: Relativity and Cosmology — The Hong Kong University of Science and Technology.
  coursera.org/verify/SV3TLHHE9AK9
- From the Big Bang to Dark Energy — The University of Tokyo.
  coursera.org/verify/X6CU9BNMSFXZ
- Understanding Modern Physics II: Quantum Mechanics and Atoms — The Hong Kong University of Science and Technology.
  coursera.org/verify/4RCAM2E9NMUA

Awards

- Full PhD Scholarship Holder (SECIHTI, Mexico) — ESFM-IPN (2025–2029)
- First Position in Master's, COMSATS University Islamabad (2023)
- Dr Abdus Salam Merit-Based Fellowship, COMSATS University Islamabad (2022)
- Gold Medalist, GOMAL University (2021)

Volunteer Work

- Organizer of Dr.Riazuddin science festival at CUI.
- Arranged three (3) blood donation camps at CUI in collaboration with KBDO (Kids Blood Disease Organization).
- Volunteered to help flood victims by providing clothes, food, and shelter.
- Member of Physics Society, CUI.

Languages

- Urdu (native); English

References

- Prof. Ruben Cordero Elizalde, ESFM-IPN.
  rcorderoe@ipn.mx
- Prof. Tonatiuh Matos, CINVESTAV.
  tonatiuh.matos@cinvestav.mx